



Birla Institute of Technology & Science, Pilani

Pilani | Dubai | Goa | Hyderabad

ACADEMIC— UNDERGRADUATE STUDIES DIVISION

FIRST SEMESTER 2026-2027

Course Handout Part II

Date: July 31, 2026

In addition to part I (General Handout for all courses, appended to the timetable), this portion provides further specific details about the course.

Course No. : CS F407/ CS U407
Course Title : ARTIFICIAL INTELLIGENCE
Instructor-in-charge : GOPAL SINGH PHARTIYAL
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Course Description

Artificial Intelligence (AI) is a field of computer science that seeks to build systems that emulate human cognition. A primary goal of AI is to build intelligent entities that mimic human intelligence, such as problem-solving, reasoning, planning, uncertainty handling, and learning. The course introduces students to a large collection of algorithms and techniques for building these intelligent entities. By the end of the course, the students should have a general knowledge of the field of AI. They should be able to recognize when and how to use AI techniques to solve problems. The students should also be able to evaluate new techniques they encounter.

This course covers the issues and techniques involved in creating computer systems that exhibit intelligent behavior. The following are among the topics that we will cover: AI search techniques, Game Playing, Planning, Knowledge Representation, Reasoning under Uncertainty, and Reinforcement Learning.

Text Book:

TB. Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, Pearson, 3e, Pearson

Reference Books:

R1. Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India), 2013.

R2. Stefan Edelkamp and Stefan Schroedl. Heuristic Search: Theory and Applications, Morgan Kaufmann, 2011.

LECTURE PLAN (40 lectures)

Note: The section order has been revised — Learning (formerly Section 5) is now covered immediately after the Introduction, once AI has been positioned relative to ML, DL, CV, GenAI, and intelligent agents.

Topic	Topic Details	No. of Lectures	Chapter Reference	Course Learning Outcome
Section 0 and Section 5: Course Placement and Prerequisites (6 lectures)				
P	1. Positioning AI on the Venn diagram: AI vs. ML, DL, CV, GenAI, and intelligent agents	1	Class Notes	Correctly place AI relative to ML, DL, CV, GenAI, and intelligent agents.
Machine Learning	2. Supervised learning 3. Unsupervised learning 4. Ensemble learning Reinforcement Learning	4	Ch. 18 – TB Ch 21 - TB	Differentiate supervised, unsupervised, ensemble, and reinforcement learning, and select an appropriate learning paradigm for a given task.
	5. Course Handout Components discussion	1	Handout	Planning for the semester



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Topic	Topic Details	No. of Lectures	Chapter Reference	Course Learning Outcome
Section 1: Introduction (3 lectures)				
Overview of AI	1. Introductory, Foundational, and Historical perspectives of AI	1	Ch. 1 - TB Class Notes	Describe the foundations and history of AI
The Turing Test: Variants and Milestones	2. Turing test & its importance 3. Turing Test Variants 4. Milestones	1	Ch.1 – TB + Class Notes + web resources	Explain the Turing test and its variants, and identify key milestones in the evolution of AI.
Agents, Environments, & Interactions	5. Types of Agents & Environments 6. Rational Behavior 7. Performance Measure, Environment, Actuators, Sensors (PEAS) 8. Structure of Agents	1	Ch. 2 – TB	Model a given problem as a rational agent using the PEAS framework and classify its task environment.
Section 3: Problem Solving (9 lectures)				
Search	1. Problem solving agents 2. Search algorithms 3. Uninformed search strategies o Breadth-first search o Depth-first search o Depth-limited search o Iterative deepening depth-first search 4. Informed search strategies o Greedy best-first search o A* algorithm 5. Heuristic functions 6. Time & space complexity of search algorithms 7. Local search algorithms o Hill climbing o Simulated annealing o Local beam search o Genetic Algorithms	7	Chs. 3-5 – TB	Formulate problems as state-space search; apply and compare uninformed, informed, and local search algorithms in terms of completeness, optimality, and complexity; design admissible heuristics.
Constraint Satisfaction Problems (CSPs)	8. Definition 9. Modeling real-world problems as CSP 10. General purpose heuristic 11. Constraint propagation 12. Backtracking search for CSPs 13. Local search for CSPs 14. Structure of problems	2	Ch. 6 – TB	Model real-world problems as CSPs and solve them using constraint propagation, backtracking, and local search.
Section 4: Knowledge, Reasoning & Planning (4 lectures)				



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Topic	Topic Details	No. of Lectures	Chapter Reference	Course Learning Outcome
Logic & Inference	1. Knowledge-based agents 2. Propositional logic 3. First-order logic (FOL) 4. Inference in FOL 5. Theorem proving & model checking 6. Knowledge engineering in FL	2	Ch. 7 – TB Ch. 8 – TB Ch. 9 – TB	Represent knowledge in propositional and first-order logic, and perform sound inference using theorem proving and model checking.
Planning	7. Classical planning 8. Heuristics for planning 9. Hierarchical planning 10. Planning in non-deterministic domains	2	Ch. 10 – TB Ch. 11 – TB	Construct classical and hierarchical plans for goal achievement, and extend planning to non-deterministic domains.
Section 5: Dealing with Uncertainty (8 lectures)				
Quantifying Uncertainty	1. Bayesian probability & Bayes' theorem 2. Naïve Bayes' model	4	Ch. 13 - TB	Apply Bayesian probability, Bayes' theorem, and the naïve Bayes model to quantify and reason about uncertainty.
Probabilistic Reasoning	3. Bayesian Belief Networks (BBN) 4. Semantics of BBNs	4	Ch. 14 - TB	Construct Bayesian belief networks for a given domain and interpret their semantics for probabilistic inference.
Section 6: Communicating, perceiving, and acting (6 lectures)				
Communicating, perceiving, and acting	1. Natural Language Processing 2. NL for Communication 3. Vision 4. Robotics 5. Future of AI	6	Web + Ch. 22 – TB Ch. 23 – TB Ch. 24 – TB Ch. 25 – TB Ch. 27 – TB	Outline how AI systems communicate (NLP), perceive (vision), and act (robotics), and critically discuss emerging directions and the future of AI.



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Evaluation Scheme:

Component	Duration	Weightage	Date (Time)
MidSem Test (Closed Book)	90 Mins	25%	October 08, 2026
Building Geospatial Intelligent Agents Project: Phase-wise Assignment(s) Number of Phases = 3 or more	Take Home	30%	Continuous Evaluation (submissions and Viva)
Quizzes (Open/Closed Book) Syllabus: course, class notes, and project	10 Mins each	10%	In lecture
Comprehensive Exam (Closed book)	180 Mins	35%	December 11, 2026

Assignment(s): Students would be required to self-read, learn, and build a particular aspect of a given topic of an existing or novel Intelligent System.

The domain of this “Intelligent System” will be informed in the first few weeks of the start of the course. You will also be required to design and demonstrate a working solution to the identified problem.

For this assignment, students will form a group (group-size = ~ 2 or less. Other details will be communicated in due course of time)

Viva-voce sessions will be arranged for group presentations.

Notices: All notices will be uploaded only on NALANDA.

Chamber Consultation Session: Amit Dua: Friday (3 - 4 PM). Preference will be given to prior requests that need to be sent to NALANDA to save time and avoid overlap.

Makeup Policy: To be granted only in case of serious illness or emergency.

Plagiarism Policy: If any student is found to be involved in any kind of plagiarism in any of the evaluation components, the matter will be directly reported to the Examination Committee.

NC Policy: Students who secure 10% or fewer marks will receive an NC grade. Students in the [10-15%] bracket are also likely to get NC.

Instructor-in-charge
CS F407